Can a Computer Feel the Color of Cinema?

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1. Introduction

Movie posters play a crucial role in the marketing of films, often shaping the audience's expectations and interest.
The goal of this project is **to predict the dominant color of a movie poster based on its metadata, such as genre, budget, language, the length of the title and year of release**. Since the target variable is categorical (the color of the movie poster), this is a multiclass classification goal.

2. Data

The dataset used in this study was collected through web scraping from the website HuggingFace.co, which hosts a variety of datasets including movie-related metadata. We focused on extracting data for a set of movie posters, collecting a total of 3,000 movie entries (N = 3000). For each movie, we retrieved features such as **title, release year, genre, budget, original language, and the most dominant color in the movie poster (see Figure 1 and notebook python)**.

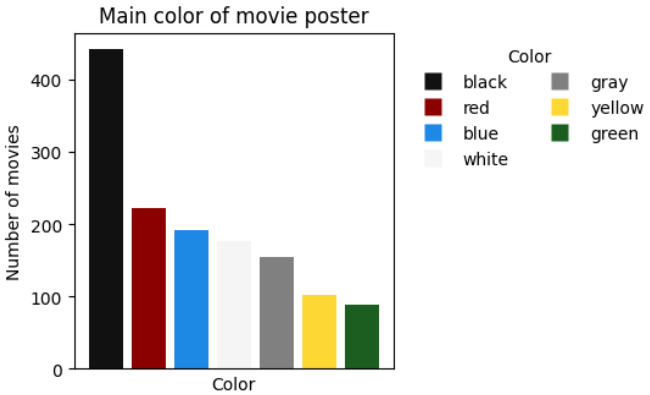

Figure 1 — Distribution of the dependent variable

3. Method

I created several new variables derived from the existing metadata:

- **logarithm of the movie budget** (to reduce scale differences between small and large productions)
- **title length**, calculated as the number of characters in each movie title
- **variables** for both *genres* and *original languages*, to transform categorical data into numerical form.

I then standardized the numerical features and removed entries with missing or zero budgets. The data was split into a **training set (75%)** and a **testing set (25%)**. I then built and evaluated three **classification models** to predict the dominant color of each movie poster:

- **Logistic Regression**
- **LASSO Logistic Regression** (L1 regularization, $\lambda = 0.1$)
- **Ridge Logistic Regression** (L2 regularization, $\lambda = 0.1$)

Since the target variable (poster color) is **multiclass and moderately unbalanced**, I used **accuracy** as the main evaluation metrics to compare model performance.

4. Results

4.1. Logistic regression

The model achieves **35.52% accuracy on the training set**, but its accuracy drops to **27.53% on the test set (see Figure 1)**. This noticeable gap suggests **high variance**. The low overall accuracy also indicates **bias**, meaning the model struggles to capture the underlying patterns effectively.

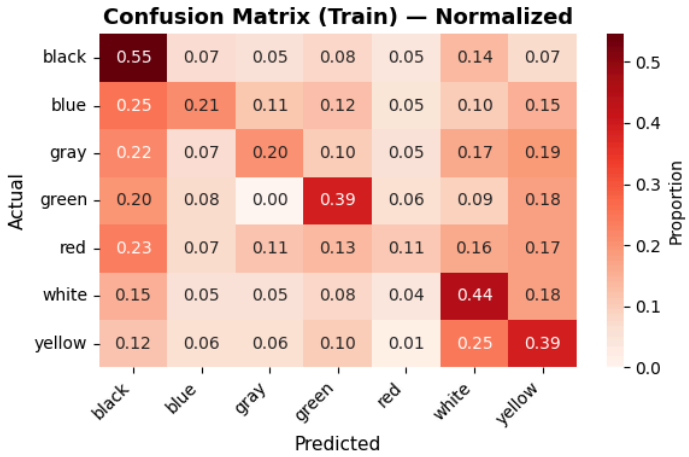

Figure 1 - Confusion matrix of the test dataset for the logistic regression model. The accuracy is 27.53%.

4.2. LASSO Logistic regression

The model achieves **35.14% accuracy on the training set**, and **34.49% on the test set**. The small difference suggests **low variance**. However, the low overall accuracy indicates **high bias**, likely due to the regularization strength of LASSO, which leads to underfitting.

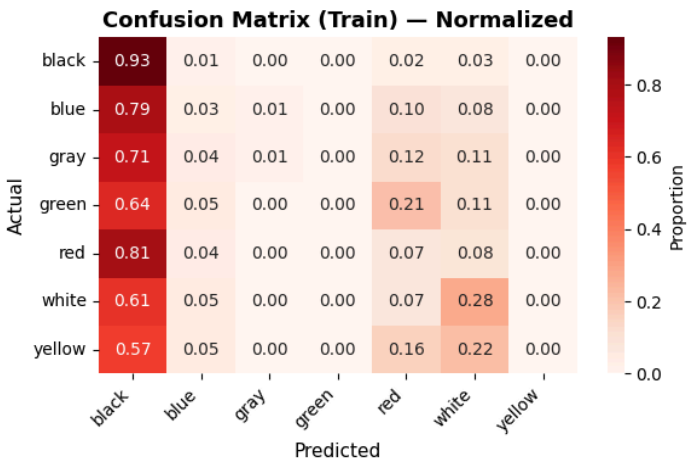

Figure 2 - Confusion matrix of the test dataset for the LASSO logistic regression model. The accuracy is 34.49%.

4.3. Ridge Logistic regression

The model achieves **35.14% accuracy on the training set**, and **34.49% on the test set**. The small difference suggests **low variance**. However, the low overall accuracy indicates **high bias**, likely due to the regularization strength of LASSO, which leads to underfitting.

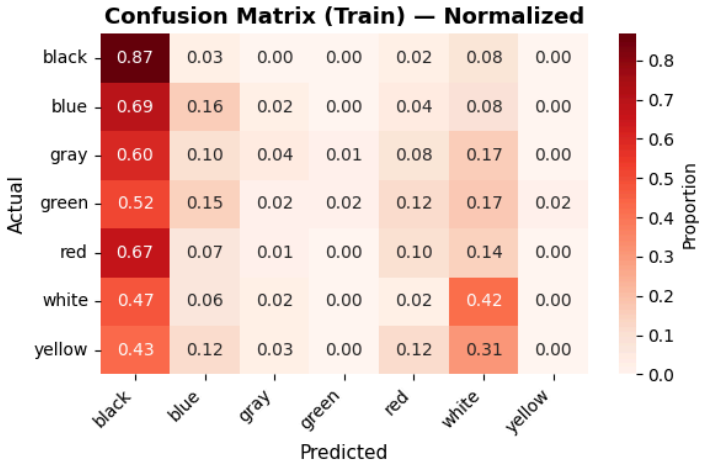

Figure 3 - Confusion matrix of the test dataset for the Ridge logistic regression model. The accuracy is 34.49%.

5. Conclusion

The best model is the Ridge regression model, with an accuracy of 37.75% on the training set and 34.78% on the test set (well above random guessing, around 16%).

This means that the Ridge model is able to correctly predict the dominant color of a movie poster in roughly one out of three cases, solely from the film's metadata such as its genre, budget, language, and year of release.

While this level of accuracy remains moderate, it demonstrates that non-visual features still contain partial but measurable information about the visual identity of a movie poster, making Ridge regression the most effective and reliable approach among all tested models.

6. References

Data collected via web scraping from *HuggingFace.co*, page:
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Fagerholm, Cecilia. "The use of colour in movie poster design: An analysis of four genres." (2009).
https://www.theseus.fi/bitstream/handle/10024/2708/Fagerholm_Cecilia.pdf?sequence=1&isAllowed=y

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